FORMULA SHEET · 02

PFL-AI Formula Sheet

PCI AI Project Finance Leader — the complete quantitative reference

NPV · IRR · WACC · CFADS · DSCR · LLCR · PLCR

PCI AI · PROJECT CONTROLS INSTITUTE GLOBAL

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AI proposes. The professional disposes.

How to use this sheet

Every formula the PFL-AI Body of Knowledge teaches, in one place, with the domain where it is worked in full.

Three columns matter. **Formula** is the expression. **Meaning** is what it answers. **Watch for** is the mistake that formula attracts in practice.

This sheet is a study aid. The examination specification, including whether any formula reference is provided, is confirmed by the Institute for each published examination form.

Conventions. Cash flows are period-end unless stated. Rates are per period and must match the period of the cash flows — an annual rate applied to semi-annual flows is the most common error in project finance modelling. Ratios to 2 dp. Currency in USD unless stated.

> **The consistency rule.** Discount rate, cash-flow period and compounding frequency must agree. If debt service is semi-annual, the discount rate is the semi-annual rate, and `n` counts half-years. Most coverage-ratio errors are period-mismatch errors, not formula errors.

1. Notation

SYMBOL	MEANING	UNIT
`CF_t`	Cash flow in period `t`	currency
`r`, `n`	Discount rate per period, number of periods	ratio, count
`DF`	Discount factor	ratio
`AF`	Annuity factor	ratio
`NPV`, `IRR`	Net present value, internal rate of return	currency, ratio
`MIRR`	Modified internal rate of return	ratio
`CFADS`	Cash flow available for debt service	currency
`DS`	Debt service — interest plus scheduled principal	currency
`DSCR`	Debt service cover ratio	ratio
`LLCR`, `PLCR`	Loan life and project life cover ratios	ratio
`DSRA`, `MRA`	Debt service and maintenance reserve accounts	currency
`D`, `E`, `V`	Debt, equity, total capital (`V = D + E`)	currency
`Rd`, `Re`	Cost of debt, cost of equity	ratio
`kd`	Discount rate used for cover ratios (typically cost of debt)	ratio
`t`	Tax rate (in cost-of-capital contexts)	ratio
`WACC`	Weighted average cost of capital	ratio
`MOIC`	Multiple on invested capital	ratio
`EBITDA`	Earnings before interest, tax, depreciation, amortisation	currency

2. Time value of money · Domain 3

FORMULA	MEANING	WATCH FOR	DOMAIN
`FV = PV(x) × (1 + r)^n`	Compounding	—	3.1
`PV(x) = FV ÷ (1 + r)^n`	Discounting	—	3.1
`DF = 1 ÷ (1 + r)^n`	Discount factor	—	3.1
`AF = [1 − (1 + r)^-n] ÷ r`	Annuity factor — PV of 1 per period for `n` periods	Applying to a growing stream	3.2
`PV(perpetuity) = CF ÷ r`	Level perpetuity	—	3.2
`PV(growing perpetuity) = CF_1 ÷ (r − g)`	Gordon growth	Valid only where `g < r`	3.2
`Effective annual rate = (1 + r/m)^m − 1`	Nominal to effective, `m` compounds per year	Comparing a nominal to an effective rate	3.3
`Real rate ≈ [(1 + nominal) ÷ (1 + inflation)] − 1`	Fisher relation	The additive approximation drifts at high inflation	3.4
`Payment = Principal ÷ AF`	Level annuity instalment	—	3.2

Annuity factor check. `AF` must equal the sum of the individual discount factors. At 8% over 5 periods, `AF` = 3.9927, and the five discount factors sum to 3.9927. If they disagree, the period convention is wrong.

3. Investment appraisal · Domain 4

FORMULA	MEANING	WATCH FOR	DOMAIN
`NPV = Σ CF_t ÷ (1 + r)^t`	Net present value	Including financing flows in an unlevered NPV	4.1
`IRR: NPV = 0`	Internal rate of return	Multiple IRRs when signs change more than once	4.1
`MIRR = (FV of inflows at reinvest rate ÷ −PV of outflows at finance rate)^(1/n) − 1`	Modified IRR	Mixing the two rates	4.1
`PI = PV(inflows) ÷ Initial investment`	Profitability index	—	4.2
`PI = 1 + NPV ÷ Initial investment`	Equivalent form — use as a check	—	4.2
`Payback = Years before recovery + (Unrecovered ÷ Cash flow in that year)`	Simple payback	Ignores everything after the payback point	4.2
`Discounted payback`	As above, on discounted flows	—	4.2
`EAV = NPV ÷ AF(r, n)`	Equivalent annual value — unequal lives	—	4.3
`Switching value = Δ% in a variable that drives NPV to zero`	Sensitivity breakeven	—	4.4

IRR pathologies — the three to know. (1) **Multiple IRRs** where the cash-flow sign changes more than once; the equation has more than one root. (2) **No IRR** where the flows never cross zero. (3) **Scale and reinvestment blindness** — IRR implicitly assumes reinvestment at the IRR itself, which is why MIRR exists and why NPV governs where the two disagree on ranking.

4. Cost of capital and structure · Domain 9

FORMULA	MEANING	WATCH FOR	DOMAIN
`WACC = g × k_d × (1 − T) + (1 − g) × k_e`	Cost of capital, in gearing form (the BoK's canonical statement)	Tax shield applied to the equity term	9.1
`g = D ÷ (D + E)`	Gearing — the `g` above	Confused with debt-to-equity	9.2
`k_e = R_f + β × (R_m − R_f)`	CAPM cost of equity	Equity beta used where asset beta is required	9.1
`Gearing = D ÷ (D + E)`	Leverage as a proportion of capital	Confused with debt-to-equity	9.2
`Debt : equity = D ÷ E`	Leverage as a ratio	60% gearing is 1.5 : 1, not 60 : 40 as a ratio	9.2
`β_asset = β_equity ÷ [1 + (1 − T) × D/E]`	Ungearing beta	—	9.1
`ICR = EBIT (or EBITDA) ÷ Interest`	Accounting cover; ignores principal entirely	Quoted as if it covered debt service	10.2
`Interest cover = EBIT ÷ Interest`	Income-based cover	EBITDA variant is a different covenant	9.2
`Leverage = Net debt ÷ EBITDA`	Leverage covenant	Gross vs net debt definition	10.2

5. Cash flow definitions · Domains 6, 14

The waterfall, in order. Each level is paid only after the one above.

ORDER	LEVEL
1	**Operating expenses**
2	**Tax**
3	**Senior debt interest**
4	**Senior debt scheduled principal**
5	**Reserve account top-ups — DSRA, MRA**
6	**Subordinated / mezzanine debt service**
7	**Distributions to equity, subject to the lock-up test**

FORMULA	MEANING	WATCH FOR	DOMAIN
`CFADS = EBITDA − Tax paid − Δ Working capital − Maintenance capex`	Cash available for debt service	Deducting interest — interest sits below CFADS	6.3
`DS = Interest + Scheduled principal`	Debt service	Excluding fees where the covenant includes them	10.2
`Cash available to equity = CFADS − DS − Reserve top-ups`	Distributable cash	—	6.4
`Lock-up: distributions blocked if DSCR < lock-up threshold`	Distribution test	Lock-up threshold differs from default threshold	10.3
`DSRA = Debt service × months ÷ 12`	Reserve, expressed as the shortfall it survives	—	10.3
`Lock-up trigger in cash = Debt service × threshold ratio`	The covenant as a number a team can act on	Covenant left as a ratio nobody can operate	10.3

CFADS is pre-financing. Interest and principal are what CFADS is measured against. Deducting interest before computing CFADS double-counts it and inflates every cover ratio. This is the most consequential definitional error in project finance.

6. Coverage ratios · Domain 10 — the flagship

FORMULA	MEANING	DOMAIN
`DSCR = CFADS ÷ DS`	Cover in a **single period**	10.2
`LLCR = PV(CFADS over loan life, at the loan rate) ÷ Outstanding debt`	Cover over the **life of the loan**	10.2
`PLCR = PV(CFADS over project life) ÷ Outstanding debt`	Cover over the **life of the project**	10.2
`Minimum DSCR`	The lowest DSCR in any period — the binding constraint	10.2
`Average DSCR`	Mean across periods — informative, never the covenant	10.2

How the three differ, and why all three exist.

RATIO	WINDOW	ANSWERS
DSCR	One period	**Can the project service debt this period?**
LLCR	Remaining loan term	**Is there enough cash over the loan's life to repay it?**
PLCR	Remaining project life	**Is there headroom beyond loan maturity — the tail?**

On the reserve term. The Body of Knowledge states `LLCR` and `PLCR` on CFADS alone. Many term sheets add cash reserves to the numerator, which raises both ratios — so state which convention a quoted figure uses, because the two are not comparable.

PLCR is normally greater than LLCR, because it discounts cash flows over a longer window against the same debt balance. The gap between them is the **tail** — the cash-generating life of the project after the debt matures. A thin tail is a credit concern even when LLCR looks comfortable, because it removes the lender's room to reschedule.

Debt sizing from a target ratio

FORMULA	MEANING	DOMAIN
`Max debt service = CFADS ÷ Target DSCR`	Affordable service per period	10.1
`Max debt capacity = Max debt service × AF(r, n)`	Quantum at the loan rate and tenor (level profile)	10.1
`Sculpted capacity = Σ [Service_t ÷ (1 + r)^t]`	Quantum where service is sculpted to CFADS	10.1
`Sculpted debt service_t = CFADS_t ÷ Target DSCR`	Coverage constant by construction	10.1
`Sculpted principal_t = Sculpted service_t − Interest_t`	Repayment shaped to cash flow	10.1

Sculpting versus annuity. A sculpted profile sets debt service as a constant multiple of CFADS, so DSCR is flat at the target in every period. An annuity profile sets a constant instalment, so DSCR varies with CFADS. Sculpting maximises debt capacity for a given minimum DSCR — which is why project finance uses it and corporate lending largely does not.

FORMULA	MEANING	DOMAIN
`Average debt life = Σ (Principal_t × t) ÷ Σ Principal_t`	Weighted average life	10.1
`Tail = Project life − Loan life`	Headroom beyond maturity	10.2
`Gearing achieved = Debt capacity ÷ Total project cost`	Resulting leverage	10.1

7. Equity returns · Domains 6, 15

FORMULA	MEANING	WATCH FOR	DOMAIN
`Equity IRR: NPV of equity cash flows = 0`	Return to shareholders	Computed on project rather than equity flows	6.4
`Project IRR`	Return on total project cost, pre-financing	—	4.1
`MOIC = Total distributions ÷ Total equity invested`	Money multiple	Ignores timing entirely	15.2
`Equity IRR > Project IRR when Project IRR > cost of debt`	The leverage effect	Leverage magnifies losses symmetrically	9.2
`Payback to equity`	Period to recover equity injections	—	15.2

Two IRRs, two questions. Project IRR asks whether the asset is worth building. Equity IRR asks whether the deal is worth doing. A strong equity IRR on a weak project IRR is a statement about leverage, not about the asset.

8. Construction phase · Domain 14

FORMULA	MEANING	WATCH FOR	DOMAIN
`Drawdown_t = Cost incurred_t × Debt proportion`	Pro-rata debt draw	Equity-first structures draw differently	14.1
`IDC = Σ Interest on drawn balance during construction`	Interest during construction	Omitted from project cost	14.2
`Total project cost = Construction + IDC + Fees + Working capital + Reserves`	Funding requirement	IDC and fees excluded	14.2
`Cost to complete = Total budget − Costs incurred`	Remaining requirement	Not the same as remaining committed	14.3
`Funding adequacy: Available funds ≥ Cost to complete`	The lender's construction test	—	14.3
`EAC (see PCL-AI sheet §4.2)`	Forecast construction outturn	The controls and finance forecasts must reconcile	8.2

The bridge to project controls. A lender's cost-to-complete test and a project controls EAC answer the same question from two sides. If the controls function forecasts an outturn above the funded amount, the funding adequacy test has already failed — usually before anyone in the finance team has been told.

9. Risk and sensitivity · Domain 11

FORMULA	MEANING	DOMAIN
`EMV = Probability × Impact`	Expected monetary value	11.2
`Switching value = Δ% to drive NPV or minimum DSCR to its limit`	Breakeven sensitivity	11.3
`Sensitivity = Δ Output ÷ Δ Input`	Elasticity of the metric to a driver	11.3
`Downside case: minimum DSCR ≥ 1.00`	Base survival test	11.3
`Breakeven tariff / volume`	Price or throughput at which DSCR hits the covenant	7.2

The standard sensitivity set. Capex overrun, delay to commercial operation, revenue or tariff reduction, opex increase, interest-rate movement, and availability or throughput shortfall — each run singly, then in a combined downside case.

10. Worked micro-examples

Each is internally consistent and reproducible from this sheet.

Appraisal. Cash flows −1,000 · 300 · 400 · 500 · 400, discount rate 10%.

- `NPV` = **252.17**
- `IRR` = **20.50%** (NPV at that rate is zero)
- `MIRR` at 8% finance / 10% reinvestment = **16.36%**
- `PI` = 1,252.17 ÷ 1,000 = **1.252** = 1 + 252.17 ÷ 1,000 — the two forms agree
- Payback = 2 + 300 ÷ 500 = **2.60 years**

Discount and annuity factors at 8% over 5 periods: `DF` = **0.6806** · `AF` = **3.9927**

DSCR. CFADS 42,000,000 · principal 18,000,000 · interest 9,500,000.

- `DS` = 27,500,000 → `DSCR` = 42,000,000 ÷ 27,500,000 = **1.53**

LLCR and PLCR. Debt outstanding 210,000,000 · loan rate 7% · CFADS 38,000,000 for 8 remaining loan years, then 36,000,000 for a further 7 project years.

- PV of CFADS over loan life = 226,909,000 → `LLCR` = 226,909,000 ÷ 210,000,000 = **1.08**
- PV of CFADS over project life = 339,828,000 → `PLCR` = 339,828,000 ÷ 210,000,000 = **1.62**
- The gap is the seven-year tail. `PLCR` > `LLCR`, as it must be wherever a tail exists.
- Add a 12,000,000 DSRA to each numerator, as many term sheets do, and the pair reads **1.14** and **1.68**. Same project, different convention — which is why the convention has to be stated.

WACC. `E` 400,000,000 · `D` 600,000,000 · `Re` 13.5% · `Rd` 7.0% · tax 25%.

- Gearing = 600 ÷ 1,000 = **60%** (debt : equity = 1.5 : 1)
- `WACC` = 0.40 × 13.5% + 0.60 × 7.0% × 0.75 = 5.40% + 3.15% = **8.55%**

Debt sizing. Target DSCR 1.35 · `kd` 7% · CFADS 40.0, 42.0, 44.0, 43.0, 41.0 million over five years.

- Capacity each year = CFADS ÷ 1.35 → 29.63, 31.11, 32.59, 31.85, 30.37 million
- Debt capacity = PV of that stream at 7% = **127,423,000**
- Check: servicing exactly that capacity returns a DSCR of **1.35 in every year** — the definition of a sculpted profile.

11. The ten formulas most often misapplied

1. **Interest deducted before CFADS.** CFADS is pre-financing. Deducting interest double-counts it and inflates every cover ratio.
2. **Period mismatch.** An annual rate applied to semi-annual debt service. Rate, period and compounding must agree.
3. **LLCR and PLCR treated as interchangeable.** They discount over different windows against the same debt. The gap is the tail, and the tail is a credit question.
4. **Average DSCR quoted as the covenant.** The **minimum** DSCR is the binding constraint. An average conceals the one period that breaches.
5. **IRR ranked above NPV on mutually exclusive projects.** IRR assumes reinvestment at itself and is blind to scale. NPV governs.
6. **Multiple IRRs unnoticed.** More than one sign change means more than one root. Check the sign pattern before quoting an IRR.
7. **Gearing confused with debt-to-equity.** 60% gearing is 1.5 : 1.
8. **Tax shield applied to the equity term of WACC.** `(1 − t)` attaches to the cost of debt only.
9. **IDC and fees omitted from total project cost.** The funding requirement is larger than the construction contract, often materially.
10. **Equity IRR presented as project performance.** It is a statement about leverage as much as about the asset. Quote both.

12. Cross-references

FOR THE FULL TREATMENT	SEE
Every formula worked with figures and exercises	**The PFL-AI Body of Knowledge, at the domain cited**
Coverage ratios, sculpting and covenants in depth	**PFL-AI Domain 10 — the quantitative flagship**
The construction-phase bridge to project controls	**PFL-AI Domain 8 and Domain 14**
Cost forecasting that feeds the funding adequacy test	**PCL-AI Formula Sheet, §4.2**

Website — https://projectcontrolsinstitute.org